

SL Math School: Random Growth Models, etc.

Tutorial 2: Brownian Motion II

1 Prologue

1.1 Strong Markov Property for Brownian Motion

Let $B(t)$ be a standard Brownian Motion and define the filtration $\mathcal{F}_t$ as

$$\mathcal{F}_t = \sigma(B_s \mid s \leq t).$$

We say that an $\mathbb{R}_{\geq 0}$ -valued random variable T is a stopping time if, for all $t \geq 0$,

$$\{T \leq t\} \in \mathcal{F}_t.$$

Strong Markov Property of Brownian Motion: If T is a stopping time such that $T < \infty$ almost surely, then the process $\mathbb{1}_{T < \infty}(B_{T+t} - B_T)$ is equal in distribution to a standard Brownian Motion independent of $\mathcal{F}_T$.

2 Tutorial 2: Problems

2.1 Core Problems

Problem 1 (Running Maximum of a Brownian Motion + Bridge). Let $S_t = \max_{0 \leq s \leq t} B_s$. Using an appropriate stopping time, show that for $b \leq a$ and $a \geq 0$

$$\mathbb{P}(S_t \geq a \text{ and } B_t \leq b) = \mathbb{P}(B_t \geq 2a - b).$$

Conclude that for all $a \geq 0$

$$\mathbb{P}(S_t \geq a) = \mathbb{P}(|B_t| \geq a) = 2\mathbb{P}(B_t \geq a).$$

and also that

$$\mathbb{P}(S_t \geq a \mid B_t = b) = \exp\left(-\frac{2a(a-b)}{t}\right).$$

Remark: The above is conditioned on a probability 0 event, so you should interpret the above conditioning in a limiting sense. In general, we will refer to a Brownian Motion on the interval $[0, t]$ starting at a and conditioned to end at b as a Brownian Bridge. This is equivalent to conditioning on the value of the very first Gaussian in our dyadic construction.

Problem 2 (Localization of Brownian Motion Minus a Parabola). Let $B(x)$ be a two-sided Brownian Motion so that $B(0) = 0$, and define $X(x) = B(x) - x^2$. Show for $R > 3$ that there exist constants $c, d > 0$ so that

$$\mathbb{P}\left(\sup_{|x| \geq R} X(x) \geq -1\right) \leq ce^{-dR^3}.$$

Conclude that

$$\mathbb{P}\left(\sup_{|x| \geq R} X(x) \geq \sup_{|x| \leq 3} X(x)\right) \leq ce^{-dR^3}.$$

Hint: Partition $\mathbb{R} \setminus [-R, R]$ into intervals of length 1, and apply the result of the previous problem on each interval.

2.2 Bonus Problems

Problem 3 (Brownian Bridge to Motion Comparison). Let $(B_u)_{0 \leq u \leq t}$ be a standard Brownian motion started at 0, and let $(W_u)_{0 \leq u \leq t}$ be a Brownian bridge from 0 to a on $[0, t]$. Let

$$\mathbb{P}_s^W = \text{law}((W_u)_{0 \leq u \leq s}), \quad \mathbb{P}_s^B = \text{law}((B_u)_{0 \leq u \leq s}).$$

Show that, for all $0 < s < t$, $\mathbb{P}_s^W$ is absolutely continuous with respect to $\mathbb{P}_s^B$, with Radon–Nikodym derivative

$$\frac{d\mathbb{P}_s^W}{d\mathbb{P}_s^B}(B) = \frac{p_{t-s}(a - B_s)}{p_t(a)},$$

where

$$p_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)}.$$

Problem 4 (Brownian Bridge in a Narrow Tube). Fix $L > 0$ and let $(B_t)_{t \in [0, L]}$ be a Brownian Motion on $[0, L]$ conditioned so that $B(0) = B(L) = 0$ (equivalently B_t is a Brownian Bridge). Show that for $r_0 < 1$, there exist constants $c, d > 0$ so that for all $0 \leq r \leq r_0$

$$\mathbb{P} \left(\sup_{t \in [0, L]} |B_t| \leq r \right) \geq c \exp \left(dL \frac{\log r}{r^2} \right).$$

Hint: Break up the interval $[0, L]$ into an ϵ -mesh. Use the Brownian absolute continuity to bound the probability that the Brownian Motion stays near the origin at each of the points in the mesh, and use the control of the running max of a Brownian Bridge to bound the region between the mesh points.

Remark: With a more clever martingale argument, one can drop the logarithmic factor in the exponent. If you finish earlier, see if you can remove the log term.

2.3 Extra Challenge Problems

Remark: Solve Question 5 of the previous tutorial if you have not already. I have updated the hint to make 5b easier. Then once you have done that problem, do the following:

Problem 5 (n -Peak Estimates). Let $B(x)$ be a two-sided Brownian Motion with $B(0) = 0$ and let $X(x) = B(x) - x^2$. Define the event $\text{TP}_n(\epsilon)$ be the event where there are n distinct points $x_1, \dots, x_n$ so that

$$|x_i - x_j| \geq 1 \text{ for } i \neq j, \quad X(x_i) \geq \max_{x \in \mathbb{R}} X(x) - \epsilon, \quad \forall i = 1, \dots, n.$$

Show that there exist positive constants $C > 0$ so that for all $n \geq 2$ and $\epsilon > 0$ small enough,

$$\mathbb{P}(\text{TP}_n(\epsilon)) \leq C(\log 1/\epsilon)^{n/3} \epsilon^{n-1}$$

Hint: Use Problem 3 to localize to a finite interval $[-R, R]$. Then, cover R with $\mathcal{O}(R)$ unit intervals such that $[-R, R]$ is covered by just the middle region of these intervals. Then, condition on the noise in exactly one of these intervals, and apply your anti-concentration inequalities to bound the near-max probability on all of the other intervals.

Problem 6 (Brownian Bridge to Brownian Motion Comparison Continued). Let $(B(u))_{u \geq 0}$ be a standard Brownian motion. Fix $T > 0$ and define, for $0 \leq r < T$,

$$W(r) = \frac{T-r}{\sqrt{T}} B \left(\frac{r}{T-r} \right).$$

Show that W extends continuously to $r = T$ by setting $W(T) = 0$, and that $(W(r))_{0 \leq r \leq T}$ is a Brownian bridge from 0 to 0 on $[0, T]$.